

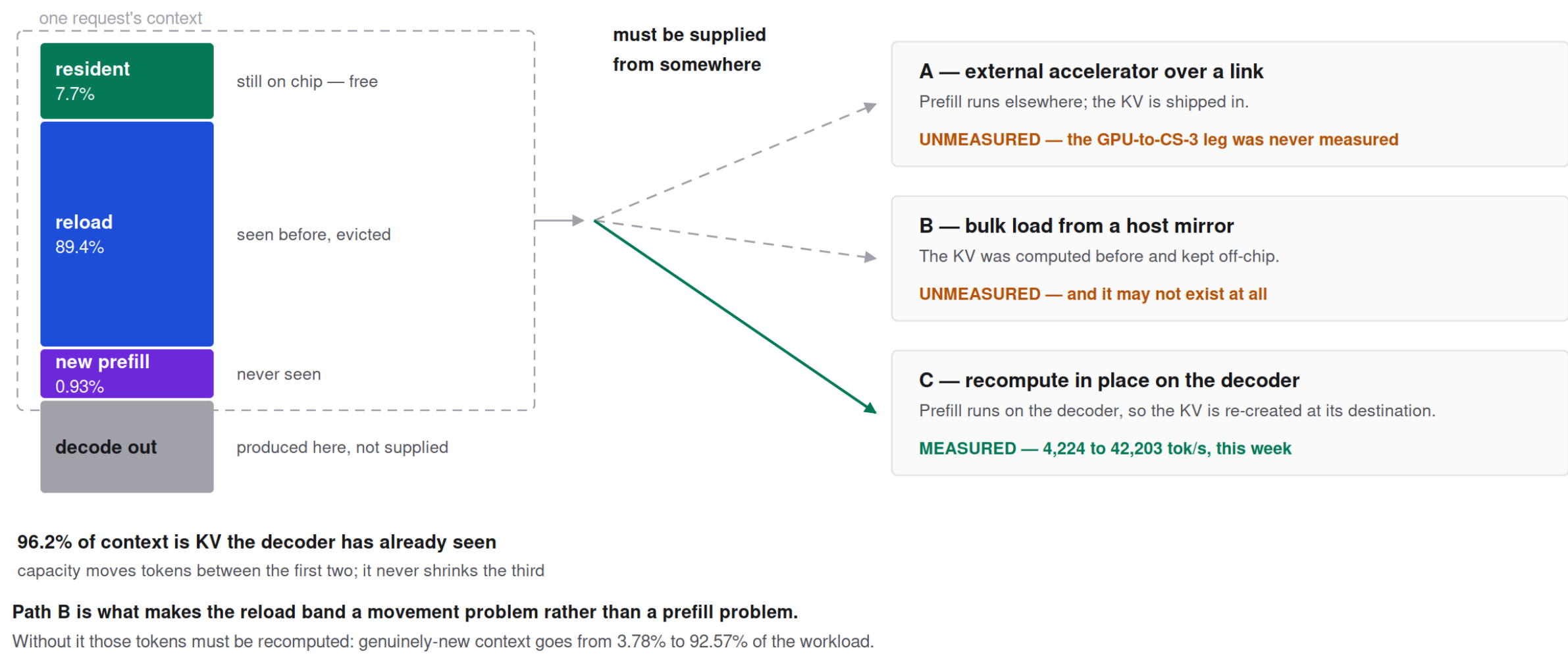
WEEKLY COLLABORATOR DISCUSSION

Where a Wafer Fits in Agentic Inference

96% of an agentic call's context is KV the decoder has already seen.
What it costs to put that back in front of the scan decides everything else.

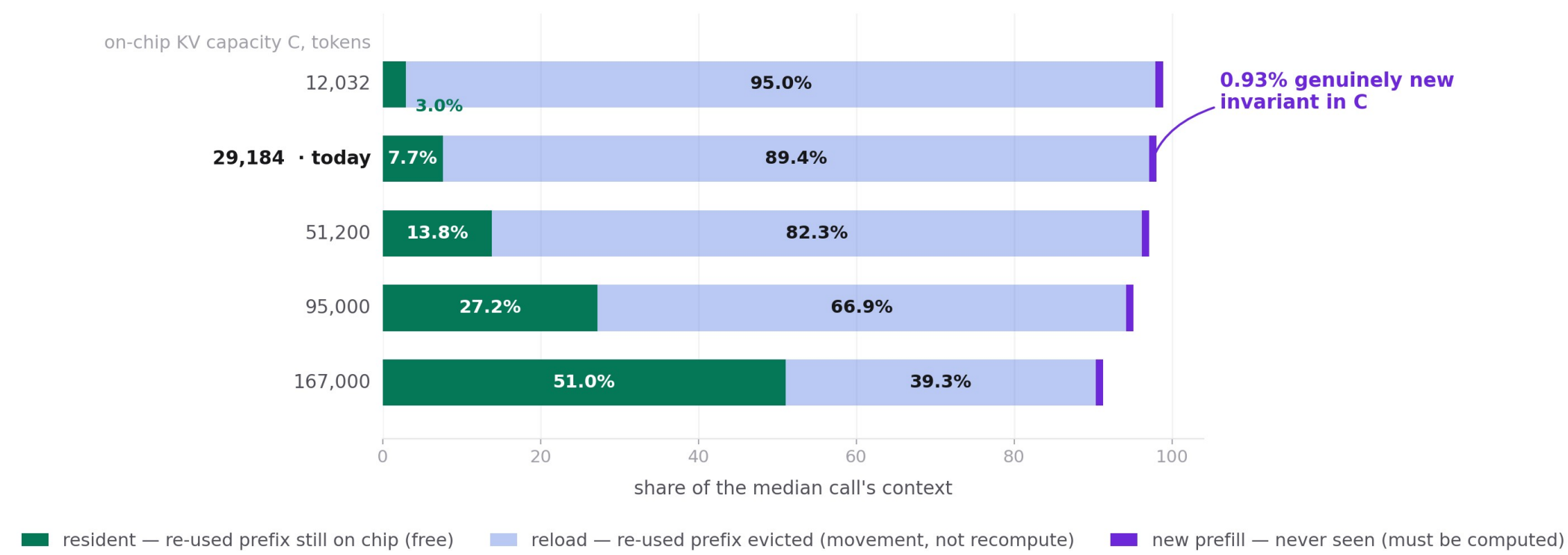
MODEL	RATES	SCOPE	DATE
Qwen3-4B	Device-measured, 750 MHz	Fits · doesn't · not yet	2026-09-06

Four kinds of token, and only three ways to supply them



Evicted prefix needs movement only, not recomputation. Only context the decoder has never seen must actually be prefilled — and those two cost an order of magnitude apart, so collapsing them hides the answer.

Only 0.9% of an agentic call is context the decoder has never seen



Claude Code, 46,650 calls, 512-token block LRU, anyblock scoring. Per-category medians, so rows do not sum to exactly 100%.

At today's 29,184-token ceiling the median call is 28,672 tokens resident, 165,376 re-used but evicted, and 1,820 genuinely new. Capacity trades the evicted band for the resident one and never touches the third.

Whether a host KV mirror exists changes the workload by 24×

Same trace, same replay, one assumption changed. Claude Code at today's 29,184-token capacity.

QUANTITY	WITH A HOST MIRROR	WITHOUT A MIRROR
Median per-call new prefill	1,820 tok	175,704 tok
Context tokens genuinely new	3.78%	92.57%
What the bottleneck becomes	Moving seen KV back	Recomputing seen KV
What a faster prefiller touches	3.78% of tokens	The whole workload
Is that path measured?	No — the link was never measured	Yes — 4,224 to 42,203 tok/s

Whether decoded KV re-enters context is a platform choice, not a law

TRACE	IS DECODED KV RE-USED?	CONSEQUENCE FOR ON-CHIP KV
Claude Code — agentic, one user, one month	~0% — thinking is stripped before the next call	Long decode consumes context and produces almost nothing keepable
ServeGen — reasoning, multi-tenant, Feb 2025	83.4% of turn pairs re-send the full previous output, reasoning included	Decoded KV is worth keeping; the 29K and 51K rungs matter there
Mooncake — chat and tool-agent, 2024	No thinking at all; output capped at 2,000 tokens	Informs the input side only — prefix sharing and arrival gaps

Three conditions — and the third is a property of the model, not the chip

CONDITION	EVIDENCE	CLASS
Decode is at least ~5% of newly prefilled tokens	Claude Code median turn sits at 63%	measured (trace)
The user waits per turn, with a large reasoning budget	5,838 decode tokens per turn, 75% hidden reasoning	measured (trace)
The decoder doubles as the prefiller	4.29–4.53× the native prefill kernel at 2K, 1.91–2.02× at 8K	measured (device)
But thinking length follows the model class	The shape evidence is frontier-model; a 4B's own agentic trace does not exist	Le's reading

Five ways the wafer loses

SITUATION	WHAT HAPPENS
No extended thinking in the turn	Parity — the per-turn advantage disappears entirely
Large new prefill, little decode	About 8× slower: first turn, repository dump, big tool output
Throughput per dollar	The GPU keeps a 15–60× margin under every lever set (modelled)
Multi-tenant aggregation	Decode-stream concurrency p50 = 1; filling S stages needs about 12·S in-turn users
Two wafers on chat-shaped traffic	Both Mooncake traces come out net negative at every prefill rate

Meeting the fit condition needs a bigger model — which needs more wafers

MODEL	WAFERS	RESIDENT CONTEXT	AGENTIC FIT
Qwen3-4B bf16	1	32K — 29,184 measured	Tool use; weak on SWE
Qwen3-8B bf16	1	24K	—
Qwen3.5-9B, dense / 3:1 hybrid	2	62K / 247K	Agentic-tuned
GPT-OSS-20B MXFP4	4	863K	Codex-class; reasoning-effort knob
Qwen3-235B-A22B bf16	12	39K — only 3K per wafer	Weight-minimum: the wrong deployment

"Does the whole request fit" is the wrong test

Capacity trades evicted KV for resident KV; it never reduces what must be prefilled.

ON-CHIP KV CAPACITY	RESIDENT	RELOAD	NEW PREFILL
12,032	2.96%	94.96%	0.93%
29,184 — today, measured	7.65%	89.44%	0.93%
51,200	13.83%	82.35%	0.93%
95,000	27.25%	66.92%	0.93%
167,000	51.04%	39.27%	0.93%

NOT YET → MEASURED · THIS WEEK'S THROUGHPUT WORK

Four stage granularities on silicon: 5.7×, paid for in context

One wafer, single request, forced prefill at 2K context, 750 MHz. Every row device-measured and gated.

LAYOUT	STAGES	CYCLES/TOKEN	TOK/S	CONTEXT CAP
A — 4 rows × 256 ² , 9 layers per block	8	101,670	7,377	29,184
L = 4 — 9 rows × 256×128	18	48,872	15,346	16,256
L = 2 — 18 rows × 256×64	36	27,348	27,424	8,128
L = 1 — 36 rows × 256×32	72	17,771	42,203	4,064

NOT YET → MEASURED · THE CEILING WAS ONE 8-BIT FIELD

Ten lines moved L=1's context ceiling from 4,064 to 12,032

The K cache is stored [dimension][position], so writing one token strides by positions-per-PE — and that stride is a signed 8-bit hardware field.

2.96x

The context ceiling at L = 1, pinned by compile sweep at 47.9 bytes per position

1 field

A signed 8-bit stride
Capped context at 127 ×
rows

10 lines

One copy, four stores
Byte-identical by construction

1,000

Records bit-identical
Free and forced paths

+0.05%

Cycle cost
Inside run-to-run variation

NOT YET → MEASURED · DECOUPLING THE TWO ROLES

Standing the roles in a column costs their compute nothing

Side by side forces attention and FFN to share a row band, so the FFN gets rows it does not need. Stacking frees the two heights.

-1.3% Attention's own work when its block is moved out of a row and into a column of its own

12,202
cycles/token measured
Three layers, gate passed

+4.1%
The strip hop
The mechanism's whole price

4.3%
Half the FFN's rows
Context-independent at every height

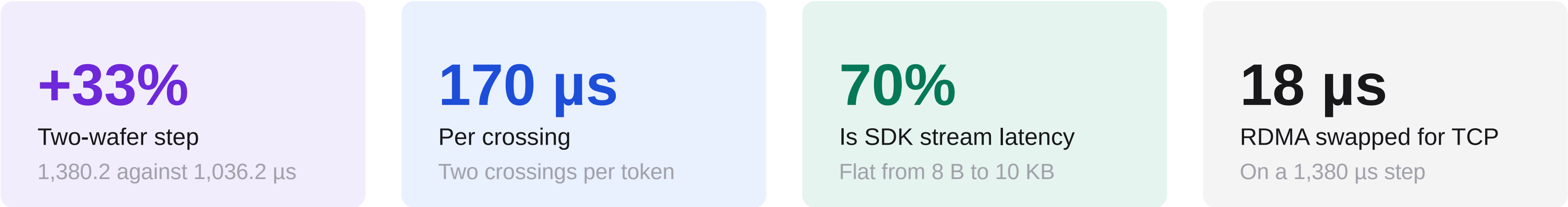
1.34×
Two lanes, projected
Same area, same ceiling

NOT YET · TWO WAFERS

The cost of a second wafer is the host, not the network

Qwen3-4B split by layer across two CS-3s, byte-exact over 4,096 steps. Each token crosses the boundary twice.

3% Share of the crossing cost that is actually the network — a faster interconnect does not fix this



The net turns on the same unmeasured fact

Claude Code, capacity 29,184 → 51,200, cost charged at 344 μs per decode token.
Actual mean is 917 decode tokens per call.

WORLD	PRICED SAVING	NET PER CALL	BREAK-EVEN
With a host mirror, any rate	Exactly zero	−0.315 s	never
No mirror, today's layout A at 8K	2.314 s	+1.998 s	6,726 decode tok
No mirror, L = 4 at 8K	1.214 s	+0.899 s	3,529 decode tok
No mirror, L = 1 at 2K	0.347 s	+0.031 s	1,008 decode tok

Boot-only routing code became a host-computed table

Appliance-compiled per-PE maxima of a 49,152 B bank. The per-PE routing setup moved to the host: 5.6 KB of boot-only code left every compute PE and 68 B of data replaced it.

REGION	BASELINE	ROUTE TABLE	FREED
Decode ATTN, 4 × 65,536 PE	39,328 B	34,080 B	−5,248 B
Decode FFN, 4 × 65,536 PE	34,944 B	30,272 B	−4,672 B
HT head / HT tail	unchanged	unchanged	0
Compile-only context ceiling	23,040 tok	30,464 tok	1.32×

DECIDE

What would move the argument

Measure first	The GPU-to-CS-3 link, and a 4B GPU curve. Both unmeasured
Design call	Is a host KV mirror part of the deployment? It moves the workload 24×
Kernel	Cut (iii): 1.71× at 24,384 context, blocked on the A1 stall
Layout	The second stacked lane: 1.34×, needs the multi-lane serpentine
Capacity	Stream KV out of idle FFN SRAM: 36,800 tokens stranded today
Workload	A same-model agentic trace on a 4B-class model